

CS3550

THE UNIVERSITY OF WARWICK

Paper Details

Paper Code: CS3550

Paper Title: Digital Forensics

Exam Period: Summer 2024

Exam Rubric

Time allowed: 2 hours

Exam Type: Standard Examination

Approved Calculators: CASIO FX82, FX83 and FX85 series

Additional Stationary

Instructions

Answer **ALL** questions from Section A and **TWO** questions from Section B.

In Section B, if more than **TWO** questions are answered, please clearly indicate which questions you wish to be marked. Otherwise, the first **TWO** questions answered in the answer booklet will be marked.

Read carefully the instructions on the answer booklet and make sure that the particulars required are entered on each answer booklet.

The exam question paper **MUST NOT** be removed from the examination venue.

Section A Answer **ALL** questions from this section.

1. (a) You have converted an RGB image to Y'CbCr color space, with

$$\mathbf{Y} = \begin{bmatrix} 20 & 21 & 22 & 23 \\ 19 & 20 & 21 & 22 \\ 19 & 19 & 20 & 22 \\ 18 & 18 & 20 & 21 \end{bmatrix}, \mathbf{Cb} = \begin{bmatrix} 21 & 24 & 19 & 17 \\ 22 & 23 & 21 & 19 \\ 23 & 21 & 20 & 24 \\ 24 & 22 & 20 & 22 \end{bmatrix}, \mathbf{Cr} = \begin{bmatrix} 22 & 23 & 24 & 21 \\ 24 & 21 & 25 & 23 \\ 22 & 24 & 26 & 23 \\ 24 & 22 & 25 & 22 \end{bmatrix}.$$

To reduce the redundancy, you are planning to perform Chroma subsampling with 4 : 2 : 0 (right) format.

Show your subsampled results for these three channels (use `floor` function to output integers). [6]

- (b) Assume you perform 2D-DCT on the 4×4 grayscale image $\mathbf{I} = \begin{bmatrix} 11 & 20 & 28 & 29 \\ 13 & 24 & 24 & 25 \\ 18 & 26 & 25 & 26 \\ 12 & 30 & 32 & 21 \end{bmatrix}$,
calculate the corresponding DC component. [5]

- (c) In image $\mathbf{I}$, given 4 pixels with values

$$\mathbf{I}(20, 14) = 70; \mathbf{I}(20, 15) = 90$$

$$\mathbf{I}(21, 14) = 100; \mathbf{I}(21, 15) = 80$$

please calculate the pixel value in $\mathbf{I}(20.4, 14.5)$ using Bilinear interpolation. [8]

- (d) Please answer the following two questions on video processing.

(i) For video encoding, there are three types of frames, i.e., I-frame, P-frame, and B-frame. What are the relationships of these three frame types? [3]

(ii) What is Group-of-Pictures (GOPs), and what are the two parameters of GOP? [3]

2. (a) There are three types of image redundancies, namely, spatial redundancy, psycho-visual redundancy and coding redundancy. Please explain what are spatial redundancy, and psycho-visual redundancy, and how to remove both redundancies for image compression. [6]

- (b) Given coefficient block C , and quantisation table Q

$$C = \begin{bmatrix} 120 & 33 & 20 & 8 \\ 40 & 30 & 18 & 4 \\ 20 & 25 & 4 & 2 \\ 10 & 15 & 2 & 1 \end{bmatrix}, Q = \begin{bmatrix} 3 & 4 & 6 & 7 \\ 4 & 5 & 8 & 10 \\ 6 & 8 & 9 & 12 \\ 7 & 10 & 12 & 20 \end{bmatrix},$$

please calculate the following:

- (i) Quantised coefficient block C' (use `round` function during quantisation); [3]
 (ii) De-quantised coefficient block C'' ; [3]
 (iii) Quantisation error in terms of SSD (sum of squared difference). [3]

- (c) Given an image $I = \begin{bmatrix} 55 & 33 & 55 & 44 & 33 \\ 33 & 55 & 55 & 55 & 33 \\ 55 & 55 & 22 & 44 & 33 \\ 55 & 55 & 22 & 44 & 55 \end{bmatrix}$, answer the following questions:

- (i) Plot the image's probability mass function (PMF); [2]
 (ii) Based on pixel probabilities in PMF, please calculate this image's entropy. [2]
 (iii) The general idea of Huffman coding is to assign short binary codewords to symbols with higher probabilities, and longer codewords to the ones with lower probabilities. Please list one set of Huffman codes for image I (Non-unique answer). [3]
 (iv) Based on the Huffman codewords in (iii), calculate the average code length. [3]

Section B Answer **TWO** questions from this section. If answering more than two questions, clearly indicate which two to mark; otherwise, only the first two answered will be marked.

3. (a) Given a 2×2 image patch $\mathbf{I} = \begin{bmatrix} 25 & 27 \\ 21 & 158 \end{bmatrix}$, answer the following questions:
- (i) Give the binary representation of $\mathbf{I}$; and show its bit-plane 0 (i.e., LSB) and bit-plane 7 (i.e., MSB) [6]
 - (ii) Describe how to create visible and invisible watermarks using bit-plane substitution methods in the pixel domain. [4]
- (b) Watermark embedding can be performed in the frequency domain, which is considered to be more robust than pixel-domain based approaches. For the best effect, the watermark is normally embedded in the perceptually significant regions of an image. What are perceptually significant regions? Why do we want to embed watermarks in these regions? [6]
- (c) Given an image f of size $M \times N$, DCT transformation can be applied to extract the frequency coefficients

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) \alpha(u) \alpha(v) \cos\left[\frac{\pi(2x+1)u}{2M}\right] \cos\left[\frac{\pi(2y+1)v}{2N}\right],$$

$$u = 0, \dots, M-1, v = 0, \dots, N-1$$

where $\alpha(u), \alpha(v)$ are scaling factors. This transformation can be simplified to

$$\mathbf{x} = \mathbf{A}\mathbf{f},$$

where $\mathbf{f}$ is the image vector such that $\mathbf{f} = [f(0, 0), f(1, 0), \dots, f(M-1, N-1)]^T$, $\mathbf{x}$ is the DCT coefficient vector $\mathbf{x} = [F(0, 0), F(1, 0), \dots, F(M-1, N-1)]^T$, and $\mathbf{A} \in \mathbb{R}^{MN \times MN}$ is the DCT transformation matrix, which is orthonormal and symmetric. Please give mathematical proof why the image's energy can be preserved under DCT transformation. [9]

4. (a) Histogram equalisation is a popular method for image contrast enhancement. Answer the following two questions:

(i) Explain why contrast can be enhanced by flattening the image histogram. [3]

(ii) Histogram matching is an extension of histogram equalization, which requires an additional reference image. Do the spatial resolution and grayscale resolution of the reference image have to be the same as the input image? Explain why. [6]

- (b) There is an image with some salt and pepper noises (i.e., dead pixels) caused by faulty

sensors. Here is an example image patch: $\mathbf{J} = \begin{bmatrix} 21 & 22 & 20 \\ 21 & 255 & 21 \\ 22 & 24 & 23 \end{bmatrix}$.

You want to enhance the image quality by using some spatial filters (i.e., removing the dead pixels). Would you use mean filter or median filter? Calculate the filtered centre pixel (of $\mathbf{J}$) based on mean and median filters respectively, and justify your choice. [4]

- (c) It is a common practise to apply filtering techniques in image processing tasks. Depending on different scenarios, the most suitable filters can be selected.

Fig. 1 demonstrates an example of both original image (left) and filtered image (right). Given two possible filters a) low-pass Gaussian filter, b) high-pass Ideal filter, which filter is more likely to be applied here? Explain your answer. [3]

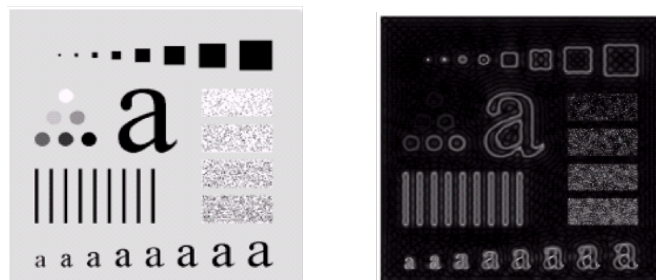


Figure 1: left: original image; right: filtered image

- (d) For a possibly spliced JPEG image, you have decided to use compression-based forensic approaches, e.g., to detect the JPEG ghosts. Answer the following questions:

(i) For JPEG ghost detection, why are pixel values normally used to compute the quantisation error, instead of DCT coefficients. [3]

(ii) Describe how to visualise the (possibly) spliced region in the target image. [3]

(iii) For a spliced JPEG image, please provide a case when JPEG ghosts cannot be detected. [3]

5. (a) Based on multiple images taken by a particular camera, Sensor Pattern Noise (SPN) can be reliably estimated. However, this approach may not work when only one noisy image (e.g., highly textured image) is available. Please explain why SPN may not be reliably estimated in this case. [6]
- (b) Given an image patch $\mathbf{I} = \begin{bmatrix} 1 & 3 & 3 \\ 2 & 1 & 4 \\ 2 & 1 & 2 \end{bmatrix}$, answer the following questions:
- (i) Compute image gradient along rows using $\nabla f_x = [-1, 1]$ [3]
 - (ii) Compute image gradient along columns using $\nabla f_y = [-1, 1]^T$ [3]
 - (iii) Based on (i) and (ii), calculate the gradient magnitude. [3]
- (c) Local Binary Pattern (LBP) is a popular feature template that can efficiently encode the texture information of an image region, and it has some good properties, e.g., it is less sensitive to illumination changes. Answer the following two questions:
- (i) Explain why LBP feature is less sensitive to illumination changes. [4]
 - (ii) Can you design a variant of LBP that is less sensitive to image rotation operation? Assume it is based on 3×3 local region. [6]
-
-